

The beginnings of industry

The invention of mechanical devices that took over much of the work of people or animals was an early step toward industrialization. The first machines were powered by water, wind, or the force of gravity, and did not need many people to operate them. Major industries would not develop until the opening of factories employing large workforces in the late 18th century.

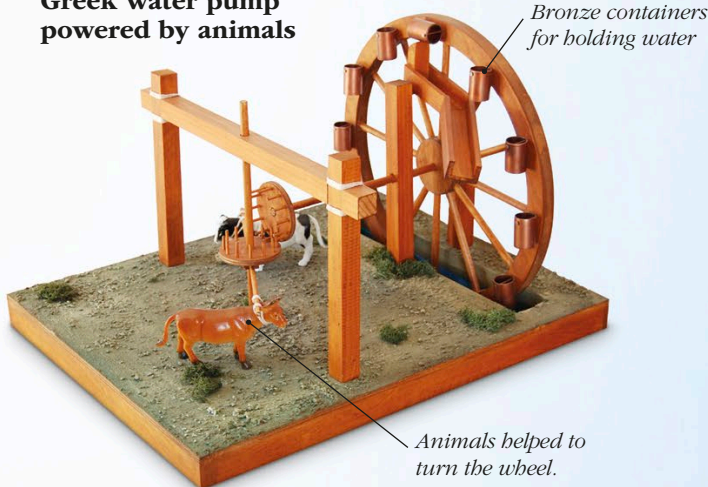
WOW!

Treadmills turned by people or animals powered some early machines. Working on a treadmill could be used as a punishment for criminals.

► LARGEST IN THE WORLD

The most famous noria were located in Hama, Syria. Seventeen can still be found there, including this one that is 72ft (22m) in diameter, the largest in the world.

Model of an ancient Greek water pump powered by animals



EARLY MACHINES

From the 4th century BCE, large wheels were invented to do work such as pumping water or driving machinery. It is uncertain where this first happened, but India, Greece, and Egypt are all possible. Some versions were powered by humans or animals. However, it was more efficient to turn the wheel using flowing water from a river or stream. These waterwheels were the first machines to turn a natural force into mechanical energy.

